

Linearization Problems for Contracting Biholomorphisms in $\mathbb{C}^n$

In local holomorphic dynamics, a central question is whether a holomorphic germ fixing the origin can be conjugated to its linear part via a local biholomorphism. In dimension one, Koenigs' theorem (1884) shows that a holomorphic germ with an attracting (or repelling) fixed point is always locally conjugate to its linear part. In higher dimensions, resonances between eigenvalues create obstructions to holomorphic linearization, leading to normal forms in the sense of Poincaré–Dulac.

These issues are closely connected to global geometry: for example, Hopf surfaces arise as compact complex surfaces obtained as quotients of $\mathbb{C}^2 \setminus \{0\}$ by a contracting biholomorphism. The dynamics of the contraction map, especially when resonances occur, have important implications for the geometry of the surface.

We then move to study quadratic Hénon maps, a class of polynomial automorphisms of $\mathbb{C}^2$, to explore linearization in the presence of resonances. This has been studied recently by David Hahn (2018). The main result shows that Hénon maps are not linearizable for resonances of orders $k = 2, \dots, 6$, while for $k \geq 7$, linearization is possible only for finitely many parameters < 1 . We present an alternative approach to his result, which reduces the problem to solving a one-variable functional equation.